• Lithosphere

22. Westerlies in southern hemisphere are stronger and persistent than in northern hemisphere. Why?

1. Southern hemisphere has less landmass as compared to northern hemisphere.

2. Coriolis force is higher in southern hemisphere as compared to northern hemisphere.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (a) : The higher speed and the greater persistence of the westerlies in the Southern Hemisphere are caused by the difference in the atmospheric pressure patterns as well as its variation from that of the Northern Hemisphere. The landmass in the southern hemisphere is comparatively less and average annual pressure decreases much more rapidly on the pole ward side of the high- pressure belt .Hence statement (a) is correct.

As far as Coriolis force is concerned, it is strongest in the polar regions and zero at the equator. At the intermediate level, it varies directly as the sine of the latitude.

23. Between India and East Asia, the navigation-time and distance can be greatly reduced by which of the following?

1. Deepening the Malacca straits between Malaysia and Indonesia.

2. Opening a new canal across the Kra isthmus between the Gulf of Siam and Andaman Sea.

Which of the statements given above is /are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (b): Navigation time and distance between India and East Asia can be reduced by opening a new canal across the Kra isthmus between the Gulf of Siam and the Andaman Sea.

24. Consider the following statements :

1. Biodiversity is normally greater in the lower latitudes as compared to the higher latitudes.

2. Along the mountain gradient, biodiversity is normally greater in lower altitudes as compared to the higher altitudes.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (c) : In mountain regions, biodiversity is generally higher in lower elevations than in higher altitudes and biodiversity is generally higher in lower latitudes than in higher latitudes.

25. The 2004 Tsunami made people realize that mangroves can serve as a reliable safety hedge against coastal calamities. How do mangroves function as a safety hedge ?

(a) The mangrove swamps separate the human settlements from the sea by a wide zone in which people neither live nor venture out.

(b) The mangroves provide both food and medicines which people are in need of after any natural disaster.

(c) The mangrove trees are tall with dense canopies and serve as an excellent shelter during a cyclone or tsunami

(d) The mangrove trees do not get uprooted by storms and tides because of their extensive roots

Ans. (d) : Mangrove forests stabilize the coastline reducing erosion from surf waves, current waves and tides. The intricate root system of mangroves also makes these forests attractive to fishes and other organisms seeking food and shelter from predators.

26. What could be the main reason/reasons for the formation of Africa and Eurasian desert belt ?

1. It is located in the sub-tropical high pressure cells.

2. It is under the influence of warm ocean currents.

Which of the statements given above is/are correct in this context?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (a): The distance from the reach of the trade winds, the location of the continent or the western edge of the site and the effect of cold currents is the main reason for the creation of the African and Eurasian desert region. Apart from this, the location in sub-tropical high- pressure cells is also an important reason.

• Hydrosphere

27. Southeast Asia has captivated the attention of global community over space and time as a geostrategically significant region. Which among the following is the most convincing explanation for this global perspective ?

(a) It was the hot theatre during the Second World War.

(b) Its location is between the Asian Powers of China and India.

(c) It was the arena of superpower confrontation during the Cold War period.

(d) Its location between the Pacific and Indian oceans and its pre-eminent maritime character.

Ans. (d) : As a geostrategic ally important area, South-East Asia attracts the attention of the global community over a long period of time and is situated between the Pacific Ocean and the Indian Ocean and has an excellent pre-eminent maritime character.

28. The surface of a lake is frozen in severe winter, but the water at its bottom is still liquid. What is the reason ?

(a) Ice is a bad conductor of heat

(b) Since the surface of the lake is at the same temperature as the air, no heat is lost.

(c) The density of water is maximum at 4^0C.

(d) None of the statements (a), (b) and (c) given above is correct.

Ans. (c): The surface of a lake is frozen in severe winter but the water at its bottom is still liquid because the density of water is maximum at 4°C. The surface water freezes at 0°C, but due to the temperature of the water below the surface being 4°C, the upper glacial surface of the lake acts as an insulator and does not allow the heat present in the water to go out.

POPULATION AND URBANIZATION

29. India is regarded as a country with Demographic Dividend. This is due to –

(a) Its high population in the age group below 15 years.

(b) Its high population in the age group of 15-64 years.

(c) Its high population in the age group above 65 years.

(d) Its high total population.

Ans. (b): India is counted as a country with the demographic dividend. The reason for this is that the population in the age group of 15-64 years is more and the number of children and old people is less.

INDIAN CONSTITUTION AND POLITY

• Citizenship, Fundamental Rights, Fundamental Duties, Directive Principles of State Policies

30. Consider the following

1. Right to education

2. Right to equal access to public service

3. Right to food

Which of the above is/are Human Right/ Human Rights under Universal Declaration of Human Rights?

(a) 1 only (b) 1 and 2 only

(c) 3 only (d) 1, 2 and 3

Ans. (d) : Right to Education (Article 26), Right to equal access to public service [Article 21] and Right to food [Article 25] comes under the Universal Declaration of Human Rights.

31. In India, if a religious sect/ community is given the status of a national minority, what special advantages it is entitled to ?

1. It can establish and administer exclusive education institutions.

2. The President of India automatically nominates a representative of the community to Lok Sabah.

3. It can derive benefits from the Prime Minister's 15 Point Programme.

Select the correct answer using the code given below

(a) 1 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

Ans. (c): Minority education has the power to reserve only up to 50% of seats for students belonging to its own community (Supreme Court Judgments on St. Stephens College Case, 1992, S.C.C. 550).

• Executive of Center and States, Emergency Provisions

32. The authorization for the withdrawal of funds from the Consolidated fund of India must come from –

(a) The President of India

(b) The Parliament of India

(c) The Prime Minister of India

(d) The Union Finance Minister

Ans. (b) : Approval of the parliament of India is Mandatory to withdraw funds from the Consolidated Fund of India. The Consolidated Fund of India is the axis of parliamentary control regarding expenditure. It is a reservoir in which all the revenue received by the Government of India is deposited and all its borrowing will be paid through it. According to Article 266 (3), funds from consolidated fund of India will be spent according to law.

33. All revenues received by the Union Government by way of taxes and other receipts for the conduct of Government business are credited to the –

(a) Contingency fund of India

(b) Public Account

(c) Consolidated fund of India

(d) Deposits and Advances Fund